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Graph-theoretic approach for $\mathcal{R}_0$ and $\mathcal{R}_e$ in continuous-time and fractional-order epidemiological models

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ABSTRACT

The study present a graph-theoretic digraph reduction method for computing the basic and effective reproduction numbers, $\mathcal{R}_0$ and $\mathcal{R}_e$, in continuous-time compartmental epidemiological models with a disease-free equilibrium (DFE). By linearizing the infected subsystem at the DFE, the dynamics are decomposed into infection and transition components represented by the Jacobian matrices F and V, yielding the classical characterization $\mathcal{R}_e$ or $\mathcal{R}_0 = \rho(FV^{-1})$, where $\rho(\cdot)$ denotes the spectral radius, and $\mathcal{R}_e$ accounting for immunity or interventions. The digraph reduction is mathematically equivalent to symbolic Gaussian elimination, avoiding explicit matrix inversion, determinant evaluation, or eigenvalue computation.

By systematically removing trivial nodes, the method reduces computational complexity, improves numerical stability, and provides tight lower bounds for reproduction numbers. The reduced digraphs reveal dominant transmission pathways and quantify compartmental contributions, enhancing interpretability and visualization. The framework naturally extends to fractional-order epidemic models, accommodating memory effects while preserving threshold consistency.

Applications to multi-compartment continuous-time and fractional-order models demonstrate that the method efficiently computes $\mathcal{R}_0$ and $\mathcal{R}_e$ in closed or semi-closed form, with substantial computational savings for high-dimensional systems. The method offers a computationally efficient and numerically stable approach, enables interpretable identification of dominant transmission pathways, is applicable to fractional-order models, and provides a generalizable framework for analyzing complex epidemiological systems.

1. Introduction

The basic reproduction number ($\mathcal{R}_0$) is a fundamental indicator in infectious disease modeling, measuring the expected number of secondary infections generated by a typical infectious individual in a fully susceptible population. Establishing whether $\mathcal{R}_0 < 1$ or $\mathcal{R}_0 > 1$ is essential before performing stability analysis, numerical simulations, or optimal control design, as this threshold dictates the feasibility of disease elimination or persistence (Diekmann et al., 1990).

The effective reproduction number ($\mathcal{R}_e$) accounts for current immunity levels or interventions. Public health strategies, including vaccination, quarantine, and treatment, critically depend on reducing $\mathcal{R}_e$ below one to control disease spread (den Driessche and Watmough, 2002).

Simple and transparent computational approaches, such as next-generation matrices, graph-theoretic reductions, or path-based decompositions, are preferred over algebraically complex methods. These techniques reduce computational errors, enhance reproducibility, and make transmission pathways interpretable (Heesterbeek et al., 2015).

Fractional-order epidemic models extend classical frameworks by naturally capturing memory effects, nonlocal interactions, and complex temporal dynamics. Fractal-fractional operators and modified Atangana–Baleanu–Caputo derivatives have demonstrated improved realism in modeling disease transmission and progression (Shah et al., 2024).

Fractal-fractional models have been applied to chronic myeloid leukemia and conjunctivitis virus infections, exhibiting richer stability and threshold behavior than integer-order models (Shah et al., 2025b). Similarly, piecewise and modified ABC derivatives have been used to analyze rotavirus transmission, breast cancer dynamics, and virus mutation processes, providing insights into disease persistence and control (Eiman et al., 2024).

Artificial intelligence and deep neural network techniques have been integrated with fractional-order modeling to address high-dimensional epidemic dynamics. For example, Khan et al. (2025) developed a fractal-fractional HIV/AIDS model combining memory effects

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with nonlocal interactions, yielding analytical results for stability and reproduction numbers.

A study by Shah et al. (2025a) applied artificial deep neural networks to a coupled chemostat system to analyze complex interactions and material recycling dynamics. The study (Ramzan et al., 2025) employed deep neural networks to approximate solutions of coupled fractional integro-differential equations, improving numerical efficiency. Either, Eiman et al. (2025a) demonstrated neural networks for fractional-order epidemic compartmental models, enhancing predictive accuracy and interpretability.

Graph-theoretic formulations provide a modular and interpretable framework for epidemic modeling, representing compartments as nodes and transitions or infections as directed edges (Segovia, 2025). These models clarify transmission chains, feedback structures, and dominant pathways (Anonymous, 2025). Graph-based models also incorporate heterogeneity and contact structure, allowing the analysis of clustering, degree distributions, and critical subpopulations (Miller, 2019).

Fractional-order epidemic models retain their next-generation matrix structure, where the infected subsystem is linearized at the disease-free equilibrium and decomposed into infection and transition components, yielding matrices F and V (Podlubny, 1998). The spectral radius of FV^{-1} defines $\mathcal{R}_0$, but stability interpretation differs due to the memory effects introduced by a fractional derivative of order α (Diethelm, 2010).

Artificial intelligence and deep neural network methods have been widely adopted to approximate solutions of complex fractional integro-differential epidemic systems (Sher et al., 2025). These methods improve numerical efficiency but often rely on classical next-generation matrices to compute $\mathcal{R}_0$ and $\mathcal{R}_e$ (Eiman et al., 2025b).

Graph-based epidemic models also support stochastic, temporal, and network-level analyses, including derivation of growth rates and effective reproduction numbers in clustered populations (Barbour and Reinert, 2022). Recent work has applied these techniques to emerging infectious diseases such as monkeypox Clade II in the USA (Meetei et al., 2025). Despite this, systematic procedures for computing $\mathcal{R}_0$ and $\mathcal{R}_e$ in complex or fractional systems remain scarce.

The present work introduces a digraph reduction methodology that is algebraically equivalent to symbolic Gaussian elimination. This approach avoids explicit matrix inverses and eigenvalue calculations while integrating naturally with both continuous-time and fractional-order epidemic models.

This work is organized as follows. In Section 2, we present a matrix-based formulation for computing $\mathcal{R}_0$ and $\mathcal{R}_e$. Section 3 introduces the graph reduction procedure, providing a systematic and computationally efficient approach. In Section 4 and Section 5, we apply this framework to both ODE and fractional-order models. Finally, Section 6 summarizes findings and outlines future directions.

2. Reproduction number $\mathcal{R}_0$ or $\mathcal{R}_e$

We begin by presenting a formula for calculating the basic reproduction number in an ODE-based disease transmission model. This approach is analogous to the derivation of the net reproductive rate for stage-structured population models, as described in Oswald et al. (2025b), de Camino-Beck et al. (2009), de Camino-Beck and Lewis (2008). Following the notation used in Oswald et al. (2025c), de Camino-Beck and Lewis (2007), Oswald et al. (2024b), we assume the ODE system possesses a disease-free equilibrium (DFE) in which all infected state variables are zero, and that the corresponding disease-free system is stable.

To analyze the system, we examine the subsystem of infected variables obtained by linearizing the ODE model at the DFE. The resulting coefficient matrix can be expressed in the form $F - V$, where F represents the contributions from new infections and V accounts for transitions among infectious compartments. As established in Li and Schneider (2002), the matrix F is entrywise nonnegative and nonzero, while V is a nonsingular M-matrix. Consequently, the next-generation matrix FV^{-1} is also nonnegative and nonzero. The basic reproduction number is then defined by $\mathcal{R}_e = \rho(FV^{-1})$, with ρ denoting the spectral radius. The DFE is locally asymptotically stable whenever $\mathcal{R}_e < 1$, and becomes unstable when $\mathcal{R}_e > 1$ (Oswald et al., 2024a; Diekmann et al., 2010; Ndendya et al., 2025d). The result below underlies the graph-theoretic approach we use to evaluate $\mathcal{R}_e$.

Theorem 1. *Suppose F is a nonnegative, nonzero matrix and V is a nonsingular M-matrix, with the property that $F - V$ is irreducible. Then the basic reproduction number satisfies $\mathcal{R}_e = \rho(FV^{-1}) > 0$. Moreover, $\mathcal{R}_e$ is equal to the reciprocal of the smallest positive solution ξ of the characteristic equation* $\det(F\xi - V) = 0$.

Proof. We assume that, the matrix FV^{-1} is nonnegative and nonzero. By Li and Schneider (2002, Proposition 4.1) and the Perron–Frobenius theorem, the principal submatrix associated with the nonzero rows of F is irreducible. Consequently, $\xi = \rho(FV^{-1}) > 0$ is the dominant positive root of the characteristic $\det(\xi I - FV^{-1}) = 0$.

Because V is nonsingular and $\lambda > 0$, this equation is equivalent to $\det(F\xi^{-1} - V) = 0$, noting that $V - F\xi^{-1}$ is a singular M-matrix. Hence ξ^{-1} corresponds to the smallest positive root. Since $\mathcal{R}_e = \rho(FV^{-1})$, the desired result follows. □

To apply Theorem 1, in the digraph reduction method (Section 3) and in the computation of $\mathcal{R}_e$ (Section 4), we examine Gaussian elimination carried out by successive row operations. Let $\mathcal{P}$ be an irreducible $n \times n$ singular M-matrix. By Berman and Plemmons (1979, Theorem 4.16), every proper principal submatrix of $\mathcal{P}$ is a nonsingular M-matrix; in particular, each diagonal entry of $\mathcal{P}$ is strictly positive.

Consider performing Gaussian elimination so that, for each $j = 1, \ldots, n-1$, the entry (j, j) is scaled to 1 and the entries (i, j) for $i > j$ are reduced to 0. At each intermediate step, all diagonal entries remain positive. At the final step, however, the (n, n) entry becomes zero because $\mathcal{P}$ is singular (Varga and Cai, 1981, Lemma 4).

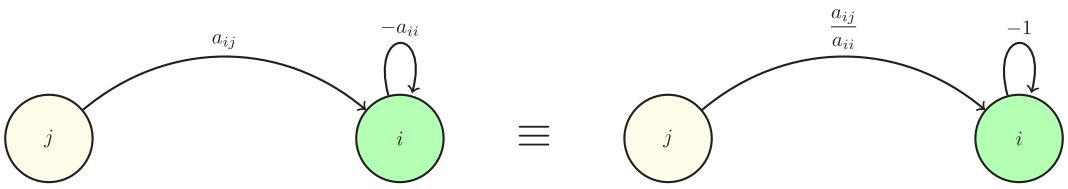
Now take $\xi = \rho(FV^{-1})$ and define $-\mathcal{P} = F\xi^{-1} - V$, so that $F\lambda^{-1} - V$ is the negative of an irreducible M-matrix. The above argument guarantees that Gaussian elimination on $F\xi^{-1} - V$ produces no zero divisors, and that all diagonal entries remain negative except at the final step, where the last diagonal entry becomes zero. Since permutation similarity preserves the M-matrix structure, the order in which columns are eliminated does not affect this property. Because $\det(F\xi^{-1} - V) = 0 \iff \det\big(E(F\xi^{-1} - V)\big) = 0, \quad \text{if } \det(E) \neq 0.$

where E is any product of elementary matrices corresponding to the row operations, Gaussian elimination yields a polynomial equation whose smallest positive root is ξ^{-1}, and hence determines $\mathcal{R}_e$ as stated in Theorem 1.

We now describe the associated digraph interpretation. For any real $n \times n$ matrix $A = (a_{ij})$, one may associate a labeled directed graph $\mathcal{D}(\mathcal{P})$ with nodes $1, \ldots, n$, where an arc $j \to i$ is present precisely when $a_{ij} \neq 0$.

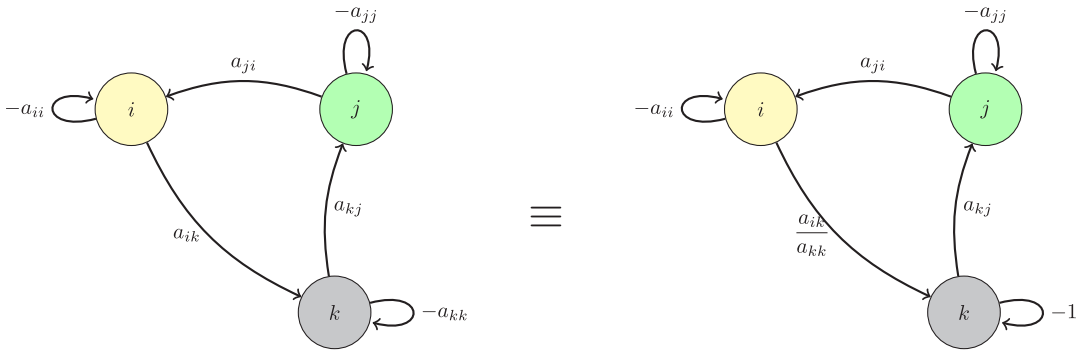
The weight of this arc is a_{ij}, and a loop is placed at node i when $a_{ii} \neq 0$. Thus $\mathcal{P}$ encodes the full structure of $\mathcal{D}(\mathcal{P})$ and conversely. This construction corresponds to the Coates digraph of $\mathcal{P}$, widely used in the engineering literature (Coates, 1959; Harary, 1962); see also (Chen, 1976, Chap. 3). (Note that in this convention, $\mathcal{P}$ is the transpose of the matrix commonly used as a weighted adjacency matrix (Heesterbeek, 2002; Harary, 1962).)

The digraph reduction rules introduced below act on the Coates digraph of $\mathcal{P}$. These rules are analogous to those in Chen (1976) and related works (Varga and Cai, 1981; Coates, 1959; Harary, 1962), and they represent elementary Gaussian elimination operations. A node of weight -1 will be referred to as a *trivial node*.



(a) Digraph representation (b) Creating a trivial node

Fig. 1. Trivial node i with self-loop and arc from j.



(a) Digraph representation (b) Creating a trivial node at node k^*



(c) Digraph representation without node k^* (d) Creating a trivial node at node j^*



(e) Creating a trivial node at node j^* (f) Digraph representation without nodes k^* and j^*

Fig. 2. Elimination of arcs through a trivial node.

3. Reduction on the digraph associated with $F\xi^{-1} - V$

3.0.1. Rule 1 (creation of a trivial node)

To normalize the loop weight $-a_{ii} < 0$ at node i to -1, divide the weight of every arc entering node i by a_{ii} (see Fig. 1(a)). Note that node j in Fig. 1(a) may also carry a loop. After applying this reduction, node i acquires a loop of weight -1, see Fig. 1(b). In terms of the determinant $\det(F\xi^{-1} - V)$, this operation corresponds to dividing row i by $a_{ii} > 0$, which multiplies the determinant by a factor of $\frac{1}{a_{ii}}$.

3.0.2. Rule 2 (elimination of arcs through a trivial node).

Suppose we are interested with node k, see Fig. 2(a), so that we can create a *trivial node* (i.e., make it has loop weight -1) and lies on a path $i \longrightarrow k \longrightarrow j$. Then the three arcs are replaced by two arcs $i \to j$ is and $j \to i$. Weight on the arc $i \to j$ is the $\frac{a_{ji}}{a_{ji}} \times a_{kj}$, see Fig. 2(c).

Once no paths pass through the trivial node k, the node can be removed entirely. Fig. 2(d) illustrates this process, where node k no longer appears in the final digraph.

If $j = i$, the weights are added to the loop, see Fig. 2(f). This is equivalent to performing an elementary Gaussian elimination operation that adds a_{ik} times row i to row j, thereby reducing the (j, i) entry to 0,

$\left(-a_{ii} + \frac{a_{ik}}{a_{kk}} \times \frac{a_{kj}}{a_{jj}} \times a_{ji} = 0\right)$. Importantly, this operation leaves the determinant unchanged.

4. Algorithm for computing $\mathcal{R}_e$ from the digraph of $F\xi^{-1} - V$

Using Theorem 1 and the two digraph reduction rules, we presented a symbolic computation method to calculate $\mathcal{R}_e$. Consider a connected dynamical system with matrix $F\xi^{-1} - V$. If this matrix is irreducible, construct the associated digraph as follows: draw an arc $j \to i$ if and only if the (i, j) entry of the matrix is nonzero. If $j = i$ and the (i, i) entry is -1, then i is called a *trivial node*.

1. In the digraph, choose a node i with a loop.
2. Use Section 3.0.1 to make node i trivial.
3. Recalculate all paths going through node i using Section 3.0.2. Node i can then be disregarded.
4. Repeat steps 1–3 until only one node remains.
5. Set the weight of the remaining loop to zero to obtain an equation for ξ. By Theorem 1, $\mathcal{R}_e$ is the reciprocal of the smallest positive root of this equation.

Note: In practice, nodes other than i can also be made trivial in step 2, but node i must be trivial before performing step 3. This algorithm is equivalent to performing symbolic Gaussian elimination on $\det(F\xi^{-1} - V) = 0$. If the digraph is not strongly connected, the algorithm can be applied to each strongly connected component separately.

5. Illustrative examples for computing $\mathcal{R}_e$ from the digraph of $F\xi^{-1} - V$

In the following Sections 5.1–5.4 all illustrative examples are presented to demonstrate how the graph–theoretic approach, together with Gaussian elimination, can be employed to obtain the basic reproduction number $\mathcal{R}_0$ and the effective reproduction number $\mathcal{R}_e$. This approach avoids the tedious and computation-heavy steps required in the classical next-generation matrix method. Instead of working through determinants, matrix inverses, or eigenvalue calculations, the examples are designed to present a simpler and more intuitive route to the same result, making the computation far less exhausting while preserving full accuracy. The values of $\mathcal{R}_0$ and $\mathcal{R}_e$ derived through this streamlined technique coincide exactly with those as established in the referenced study, showing that the graph–theoretic formulation offers an equivalent yet more accessible procedure for reaching the reproduction numbers.

5.1. Monkeypox transmission dynamics model

As simple illustrative example of Shanta and Akbar (2025), the model consists of the compartments $S(t)$, $E(t)$, $I(t)$, $Q(t)$, and $R(t)$. Susceptible individuals $S(t)$ become infected through contact with infectious individuals at rate λ. Exposed individuals $E(t)$, who are infected but not yet infectious, progress to the infectious class at rate κ, while a fraction enters early isolation at rate ρ. Infectious individuals $I(t)$ undergo quarantine at rate ϵ and may die from infection at rate δ. Quarantined individuals $Q(t)$ recover at rate τ, joining the recovered class $R(t)$ as illustrated in Fig. 3. Natural birth and death rates are neglected, so the total population is assumed constant to focus solely on the core Mpox transmission dynamics.

$$\begin{cases} \frac{dS}{dt} = -\lambda S(t) I(t), \\ \frac{dE}{dt} = \lambda S(t) I(t) - (\kappa + \rho) E(t), \\ \frac{dI}{dt} = \kappa E(t) - (\epsilon + \delta) I(t), \\ \frac{dQ}{dt} = \epsilon I(t) - \tau Q(t) + \rho E(t), \\ \frac{dR}{dt} = \tau Q(t). \end{cases} \tag{1}$$

Let us consider the infectious subsystem with $X = (E, I, Q)^\top$. At the disease-free equilibrium (DFE), we have $(S, E, I, Q, R) = (1, 0, 0, 0, 0)$ (see Fig. 4). As discussed in Sections 3 and 4, our analysis of the model system (1) begins by separating the new infection terms from the remaining population dynamics in accordance with the $\mathbb{GT}$ framework. In this formulation, $\mathbb{F}$ is a 3×3 matrix and $\mathbb{V}$ a non-singular 3×3 matrix, each obtained from the appropriate partial derivatives and evaluated at the system's equilibrium. The system can thus be written as $\dot{X} = \mathbb{F}(X) - \mathbb{V}(X)$, with

$$\mathbb{F} = \begin{pmatrix} 0 & \lambda & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{and} \quad \mathbb{V} = \begin{pmatrix} \kappa + \rho & 0 & 0 \\ -\kappa & \epsilon + \delta & 0 \\ -\epsilon & \tau & -\rho \end{pmatrix}.$$

Such that,

$$\mathbb{F}\xi^{-1} - \mathbb{V} = \begin{pmatrix} -(\kappa + \rho) & \lambda\xi^{-1} & 0 \\ \kappa & -(\epsilon + \delta) & 0 \\ \epsilon & -\tau & \rho \end{pmatrix}.$$

Creating a trivial node at node Q, we normalize the loop weight ρ so that the corresponding diagonal entry becomes -1. Specifically, if the quantity to be normalized satisfies $\rho > 0$, then we divide the entire row by $-\rho$; whereas if $\rho < 0$, we divide the row by ρ so that the resulting diagonal entry remains negative. Under this normalization, the row takes the form: $-\frac{\epsilon}{\rho} \quad \frac{\tau}{\rho} \quad -1$ (see Figs. 5 and 6).

$$\mathbb{F}\xi^{-1} - \mathbb{V} = \begin{pmatrix} -(\kappa + \rho) & \lambda\xi^{-1} & 0 \\ \kappa & -(\epsilon + \delta) & 0 \\ -\frac{\epsilon}{\rho} & \frac{\tau}{\rho} & -1 \end{pmatrix}$$

$$\implies \mathbb{F}\xi^{-1} - \mathbb{V} = \begin{pmatrix} -(\kappa + \rho) & \lambda\xi^{-1} \\ \kappa & -(\epsilon + \delta) \end{pmatrix}.$$

$$\mathbb{F}\xi^{-1} - \mathbb{V} = \begin{pmatrix} -(\kappa + \rho) & \lambda\xi^{-1} \\ \frac{\kappa}{\epsilon + \delta} & -1 \end{pmatrix}. \implies \mathbb{F}\xi^{-1} - \mathbb{V} = \left(-(\kappa + \rho) + \lambda\xi^{-1} \times \frac{\kappa}{\epsilon + \delta}\right).$$

$$\det(\mathbb{F}\xi^{-1} - \mathbb{V}) = \left|-(\kappa + \rho) + \lambda\xi^{-1} \times \frac{\kappa}{\epsilon + \delta}\right| = 0.$$

Now solving for ξ, take $-(\kappa + \rho) + \lambda\xi^{-1} \times \frac{\kappa}{\epsilon + \delta} = 0$, $\implies$ $\xi = \frac{\kappa\lambda}{(\epsilon + \delta)(\kappa + \rho)}$.

The solution of ξ is the expression of $\mathcal{R}_0$, such that $\mathcal{R}_0 = \frac{\kappa\lambda}{(\epsilon + \delta)(\kappa + \rho)}$

5.2. M. bovis transmission dynamics model

Consider the *M. bovis* model presented in Oswald et al. (2025a), that describes *M. bovis* transmission between humans and cattle. Humans $N_h(t)$ are in S_{nh}, S_{eh}, E_h, I_h, R_h, with recruitment π_h, education ψ (protection θ), infection forces $\varphi_1 = (1 - \theta)\left(a_1 \frac{I_c}{N_h} + a_2 \frac{I_h}{N_h}\right)$, $\varphi_2 = a\theta \frac{I_c}{N_h}$, progression/recovery ρ_h, natural/death rates μ_h, α_h, and immunity loss τ_h.

Cattle $N_c(t)$ are in S_{nv}, S_v, E_c, I_c, recruited at π_c, vaccinated at ω (efficacy λ), with infection forces $\sigma_1 = (1 - \lambda)\left(\varepsilon_1 \frac{I_h}{N_c} + \varepsilon_2 \frac{I_c}{N_c}\right)$, and $\sigma_2 = \varepsilon\lambda \frac{I_h}{N_c}$. Interactions occur via all direct and indirect pathways.

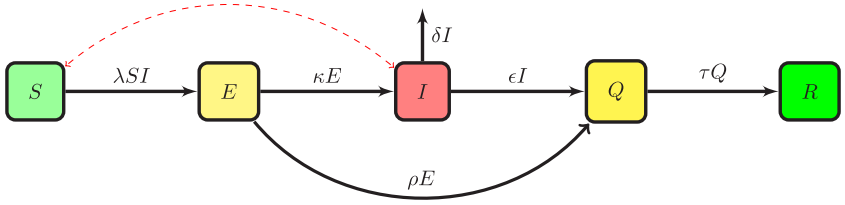


Fig. 3. Schematic diagram depicting monkeypox transmission pathways.

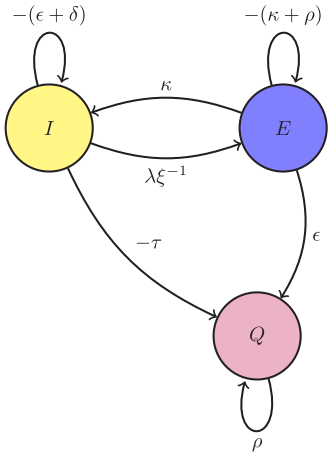


Fig. 4. Digraph representation of $\mathbb{F}\xi^{-1} - \mathbb{V}$.

$$\begin{cases} \frac{dS_{nh}}{dt} &= \pi_h - (\mu_h + \psi + \varphi_1)S_{nh}, \\ \frac{dS_{eh}}{dt} &= \psi S_{nh} - (\mu_h + \varphi_2)S_{eh}, \\ \frac{dE_h}{dt} &= \varphi_1 S_{nh} + \varphi_2 S_{eh} - (\mu_h + \gamma_h)E_h + \tau_h R_h, \\ \frac{dI_h}{dt} &= \gamma_h E_h - (\mu_h + \alpha_h + \rho_h)I_h, \\ \frac{dR_h}{dt} &= \rho_h I_h - (\mu_h + \tau_h)R_h, \\ \frac{dS_{nv}}{dt} &= \pi_c - (\mu_c + \omega + \sigma_1)S_{nv}, \\ \frac{dS_v}{dt} &= \omega S_{nv} - (\mu_c + \sigma_2)S_v, \\ \frac{dE_c}{dt} &= \sigma_1 S_{nv} + \sigma_2 S_v - (\mu_c + \gamma_c)E_c, \\ \frac{dI_c}{dt} &= \gamma_c E_c - (\mu_c + \alpha_c)I_c. \end{cases} \tag{2}$$

Now, Let E_0 be disease-free equilibrium point. Therefore, the disease free equilibrium point (DFE) denoted by E_0 can be expressed as $E_0 = \left[\frac{\pi_h}{\mu_h + \psi}, \frac{\psi}{\mu_h}\left(\frac{\pi_h}{\psi + \mu_h}\right), 0, 0, 0, \frac{\pi_c}{\mu_c + \omega}, \frac{\omega}{\mu_c}\left(\frac{\pi_c}{\omega + \mu_c}\right), 0, 0\right]$.

Following the guidelines in Section 3 and the procedure detailed in Section 4, we analyze the model system Eq. (2) by first isolating the terms associated with new infections from those governing the remaining population dynamics, as required by the $\mathbb{GT}$ framework. In this formulation, $\mathcal{K}$ denotes an $n \times n$ matrix, while $\mathcal{B}$ is a non-singular $m \times m$ matrix. Both matrices are derived from the relevant partial derivatives of the system and evaluated at the equilibrium point. Their explicit representations are provided below.

$$\mathcal{K} = \left[\frac{\partial \mathcal{K}_i}{\partial(E_h, I_h, E_c, I_c)}\right]_{E_0} \quad \text{and} \quad \mathcal{B} = \left[\frac{\partial \mathcal{B}_i}{\partial(E_h, I_h, E_c, I_c)}\right]_{E_0} \tag{3}$$

$$\mathcal{K} = \begin{bmatrix} 0 & \frac{a_2(1-\theta)\mu_h}{\psi + \mu_h} & 0 & \frac{a_1(1-\theta)\mu_h + a\psi\theta}{\psi + \mu_h} \\ 0 & 0 & 0 & 0 \\ 0 & \frac{\varepsilon_1(1-\lambda)\mu_c + \varepsilon\omega\lambda}{\omega + \mu_c} & 0 & \frac{\varepsilon_2(1-\lambda)\mu_c}{\omega + \mu_c} \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

and

$$\mathcal{B} = \begin{bmatrix} (\mu_h + \gamma_h) & 0 & 0 & 0 \\ -\gamma_h & (\mu_h + \alpha_h + \rho_h) & 0 & 0 \\ 0 & 0 & (\mu_c + \gamma_c) & 0 \\ 0 & 0 & -\gamma_c & (\mu_c + \alpha_c) \end{bmatrix}.$$

From which,

$$\mathcal{K}\xi^{-1} - \mathcal{B} = \begin{bmatrix} -(\mu_h + \gamma_h) & \frac{a_2(1-\theta)\mu_h}{\psi + \mu_h}\xi^{-1} & 0 & \frac{a_1(1-\theta)\mu_h + a\psi\theta}{\psi + \mu_h}\xi^{-1} \\ \gamma_h & -(\mu_h + \alpha_h + \rho_h) & 0 & 0 \\ 0 & \frac{\varepsilon_1(1-\lambda)\mu_c + \varepsilon\omega\lambda}{\omega + \mu_c}\xi^{-1} & -(\mu_c + \gamma_c) & \frac{\varepsilon_2(1-\lambda)\mu_c}{\omega + \mu_c}\xi^{-1} \\ 0 & 0 & \gamma_c & -(\mu_c + \alpha_c) \end{bmatrix},$$

5.2.1. *Illustration of the matrix and digraph relationships in the model*

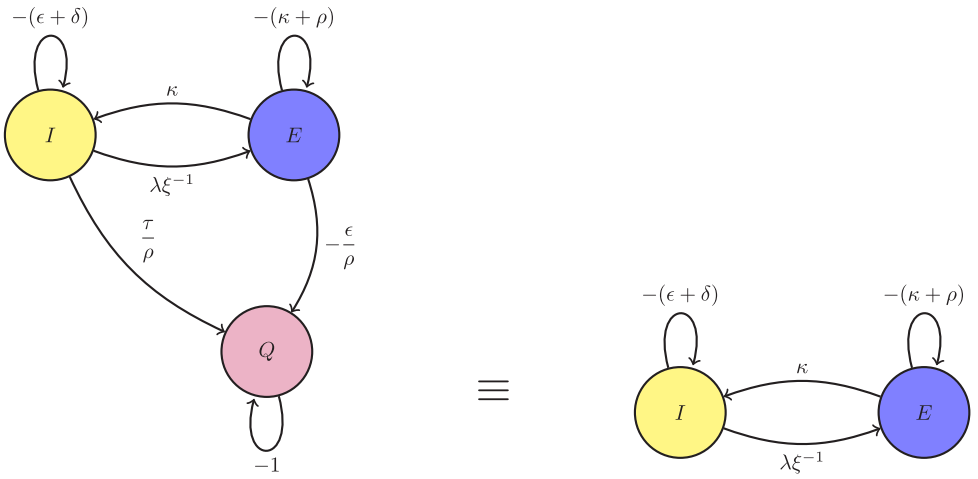
The matrix $\mathcal{K}\xi^{-1} - \mathcal{B}$ describes the transitions between epidemiological compartments, with each entry representing the rate of movement among states such as exposed, infected, or recovered individuals. The corresponding digraph provides a visual interpretation: nodes represent compartments (e.g., E_h, I_h, E_c, and I_c for human and cattle states), while directed edges indicate transition rates. Parameters such as γ_h and $\frac{\varepsilon_1(1-\lambda)\mu_c + \varepsilon\omega\lambda}{\omega + \mu_c}$ characterize disease progression and transmission. Self-loops denote internal processes like death or recovery, whereas off-diagonal edges represent transitions between distinct states, such as movement from exposed to infectious classes, as illustrated in Fig. 8.

5.2.2. *Digraph reduction and Gaussian elimination*

Digraph reduction parallels Gaussian elimination on the matrix $\mathcal{K}\xi^{-1} - \mathcal{B}$ by systematically removing intermediate nodes, analogous to eliminating off-diagonal entries. A node becomes trivial once its self-loop is normalized to -1, typically corresponding to constant rates such as natural mortality. The reduction procedure follows two fundamental rules: as outlined in Section 3, incoming edges are normalized to create a trivial node; and as described in Section 4, all paths through that node are replaced by direct edges. For instance, the node I_c is simplified by dividing its incoming edges by $(\mu_c + \gamma_c)$ and updating the resulting connections. These structural transformations are illustrated in Figs. 9 and 10 (see Box I).

where $\mathcal{V}^* = (\mu_c + \alpha_c) - \left\{\frac{\varepsilon_2(1-\lambda)\mu_c}{(\omega + \mu_c)(\mu_c + \gamma_c)}\right\}\gamma_c\xi^{-1}$ (4)

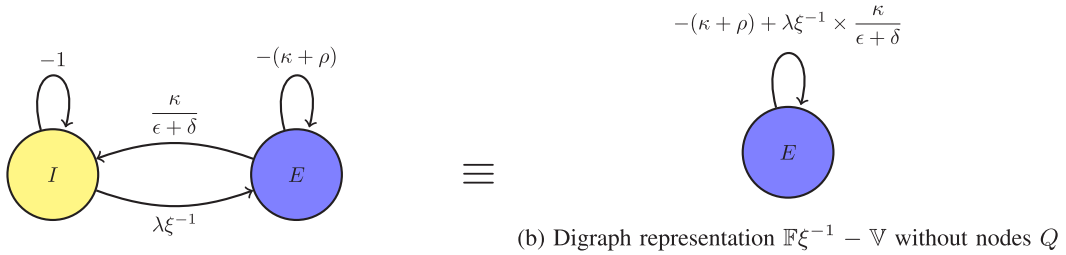
Analyzing the digraphs in Figs. 9–12 via Gaussian elimination and spectral radius evaluation provides a systematic, transparent computation of $\mathcal{R}_e$. This approach not only simplifies the derivation but also clarifies transmission dynamics by directly linking $\mathcal{R}_e$ to specific



(a) Creating a trivial node at node Q

(b) Digraph representation $\mathbb{F}\xi^{-1} - \mathbb{V}$ without node Q

Fig. 5. Elimination of arcs through a trivial node.



(a) Creating a trivial node at node I

(b) Digraph representation $\mathbb{F}\xi^{-1} - \mathbb{V}$ without nodes Q and I

Fig. 6. Elimination of arcs through a trivial node.

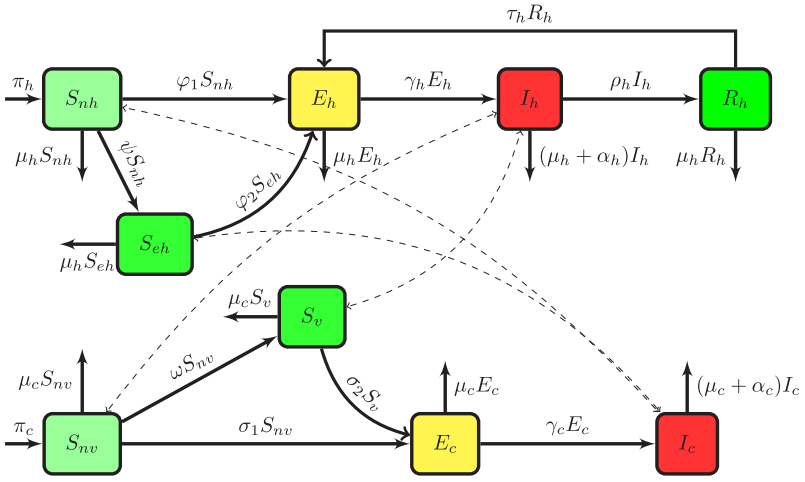


Fig. 7. Schematic diagram depicting bTB transmission pathways.

pathways within the model, see Fig. 7.

$$-(\mu_h+\gamma_h)+\frac{\gamma_h}{(\mu_h+\alpha_h+\rho_h)}\left\{\frac{a_2(1-\theta)\mu_h}{\psi+\mu_h}\xi^{-1} + \frac{(a_1(1-\theta)\mu_h+a\psi\theta)(\varepsilon_1(1-\lambda)\mu_c+\varepsilon\omega\lambda)}{\mathcal{V}^*(\omega+\mu_c)(\psi+\mu_h)(\mu_c+\gamma_c)}\gamma_c\xi^{-2}\right\}=0 \tag{5}$$

From Eq. (4), denoting $\mathcal{J}_1 = \mu_c + \alpha_c$, $\mathcal{J}_2 = \left\{\frac{\varepsilon_2(1-\lambda)\mu_c}{(\omega+\mu_c)(\mu_c+\gamma_c)}\right\}\gamma_c$, such that $\mathcal{V}^* = \mathcal{J}_1 - \mathcal{J}_2\xi^{-1}$, while from Eq. (5),

$$\mathcal{D}_1 = \mu_h+\gamma_h, \quad \mathcal{D}_2 = \frac{\gamma_h}{\mu_h+\alpha_h+\rho_h}, \quad \mathcal{C} = \frac{a_2(1-\theta)\mu_h}{\psi+\mu_h}$$

and $$\mathcal{H} = \frac{(a_1(1-\theta)\mu_h+a\psi\theta)(\varepsilon_1(1-\lambda)\mu_c+\varepsilon\omega\lambda)}{(\omega+\mu_c)(\psi+\mu_h)(\mu_c+\gamma_c)}\gamma_c,$$

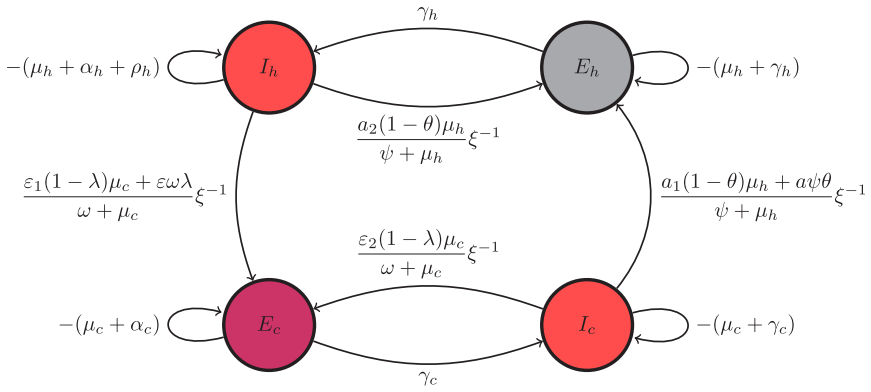


Fig. 8. Digraph representation of $\mathcal{K}\xi^{-1}-\mathcal{B}$.

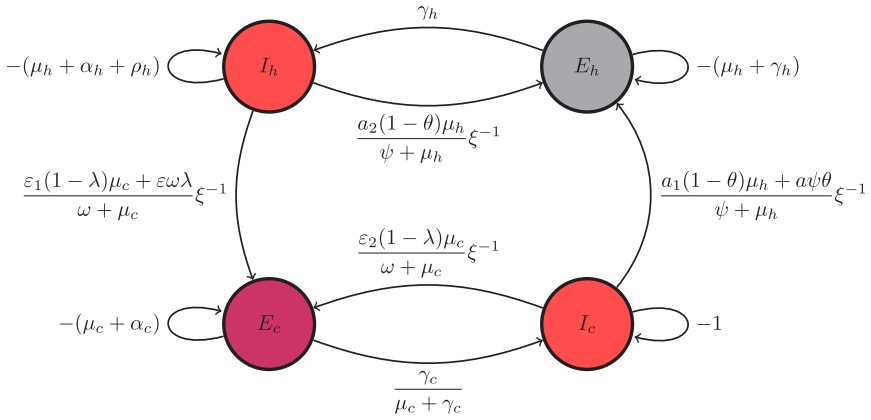


Fig. 9. Creating a trivial node at node I_c.

$$\mathcal{K}\xi^{-1}-\mathcal{B}=\begin{pmatrix} -(\mu_h+\gamma_h) & \frac{a_2(1-\theta)\mu_h}{\psi+\mu_h}\xi^{-1} & 0 & \frac{a_1(1-\theta)\mu_h+a\psi\theta}{\psi+\mu_h}\xi^{-1} \\ \gamma_h & -(\mu_h+\alpha_h+\rho_h) & 0 & 0 \\ 0 & \frac{\varepsilon_1(1-\lambda)\mu_c+\varepsilon\omega\lambda}{\omega+\mu_c}\xi^{-1} & -(\mu_c+\gamma_c) & \frac{\varepsilon_2(1-\lambda)\mu_c}{\omega+\mu_c}\xi^{-1} \\ 0 & 0 & \frac{\gamma_c}{\mu_c+\gamma_c} & -1 \end{pmatrix}$$

$$\Downarrow$$

$$\mathcal{K}\xi^{-1}-\mathcal{B}=\begin{pmatrix} -(\mu_h+\gamma_h) & \frac{a_2(1-\theta)\mu_h}{\psi+\mu_h}\xi^{-1} & \frac{\gamma_c}{\mu_c+\gamma_c}\times\frac{a_1(1-\theta)\mu_h+a\psi\theta}{\psi+\mu_h}\xi^{-1} \\ \gamma_h & -(\mu_h+\alpha_h+\rho_h) & 0 \\ 0 & \frac{\varepsilon_1(1-\lambda)\mu_c+\varepsilon\omega\lambda}{\omega+\mu_c}\xi^{-1} & -(\mu_c+\gamma_c)+\frac{\gamma_c}{\mu_c+\gamma_c}\times\frac{\varepsilon_2(1-\lambda)\mu_c}{\omega+\mu_c}\xi^{-1} \end{pmatrix}$$

Box I.

now Eq. (5) becomes

$$-D_1+D_2\left\{C\xi^{-1}+\frac{\mathcal{H}}{J_1-J_2\xi^{-1}}\xi^{-2}\right\}=0 \tag{6}$$

Simplifying Eq. (6), we get

$$-D_1J_1+D_1J_2\xi^{-1}+J_1D_2C\xi^{-1}-J_2D_2C\xi^{-2}+\mathcal{H}D_2\xi^{-2}=0 \tag{7}$$

Fo simplification, one may multiply ξ^2 on Eq. (7) and observe that

$$-D_1J_1\xi^2+(D_1J_2+J_1D_2C)\xi+(-J_2D_2C+\mathcal{H}D_2)=0 \tag{8}$$

One may realize that from Eq. (8), the effective reproduction number of the model system (2), is given by

$$\mathcal{R}_e=\frac{1}{2D_1J_1}\left(D_1J_2+J_1D_2C+\sqrt{(D_1J_2+J_1D_2C)^2+4D_1J_1(-J_2D_2C+\mathcal{H}D_2)}\right). \tag{9}$$

Since $\mathcal{R}_e$ has already been derived, then $\mathcal{R}_0$ follows immediately as a special case and does not require a separate recalculation. Specifically, in the absence of any intervention strategies, namely when the education campaign rate and vaccination rate are set to zero ($\psi=0$ and $\omega=0$), and no treatment is applied to infectious humans ($\rho_h=0$), the

expression for $\mathcal{R}_e$ reduces directly to $\mathcal{R}_0$, given by:

$$\mathcal{R}_0 = \frac{1}{2D_1J_1}\left(D_1J_2 + J_1D_2C + \sqrt{(D_1J_2 + J_1D_2C)^2 + 4D_1J_1(-J_2D_2C + HD_2)}\right), \tag{10}$$

where, $HD_2 = \frac{\gamma_c\gamma_h a_1}{\mu_c(\gamma_c+\mu_c)(\alpha_h+\mu_h)}$, $J_2D_2C = \frac{\gamma_h a_2\varepsilon_2}{(\gamma_c+\mu_c)(\alpha_h+\mu_h)}$, $J_1D_2C = \frac{(\alpha_c+\mu_c)\gamma_h a_2}{(\alpha_h+\mu_h)}$, $D_1J_2 = \frac{(\gamma_h+\mu_h)\varepsilon_2}{(\gamma_c+\mu_c)}$, and $J_1D_1 = (\alpha_c+\mu_c)(\gamma_h+\mu_h)$.

5.3. HPV transmission dynamics model

The study of Oswald et al. (2025c) stratifies the population into ages 0–9, 9–14, and > 14 to study HPV transmission under female-only vaccination for ages 9–14 and sexually active females as in Fig. 13. Births enter at rate π, children transition at rate ξ, and all groups experience mortality μ. Females are classified into S_f, V_f, E_f, I_f, Cc, and R_c, and males into S_m, E_m, I_m, and R_m. Vaccination occurs at rates σ_1 and σ_2, exposure at λ_1 and λ_2, progression at γ_1 and γ_2, recovery at ρ and κ, and cancer deaths at ν, assuming perfect efficacy and no sexual activity below 14.

$$\begin{cases}
\frac{dC}{dt} &= \pi - (\mu+\xi)C,\\
\frac{dG}{dt} &= \xi(1-\epsilon)C - \mu G,\\
\frac{dV_1}{dt} &= \sigma_1 G - (\mu+\xi)V_1,\\
\frac{dS_f}{dt} &= \xi(1-\sigma_1)G - \mu S_f - \sigma_2 S_f - (1-\sigma_2)\lambda_1 S_f,\\
\frac{dV_2}{dt} &= \sigma_2 S_f + \xi V_1 - \mu V_2,\\
\frac{dE_f}{dt} &= (1-\sigma_2)\lambda_1 S_f - (\mu+\gamma_1)E_f,\\
\frac{dI_f}{dt} &= \gamma_1 E_f - (\mu+\rho)I_f,\\
\frac{dR_f}{dt} &= \rho I_f - \mu R_f,\\
\frac{dCc}{dt} &= (1-\rho)I_f - (\mu+\nu)Cc,\\
\frac{dB}{dt} &= \epsilon\xi C - (\mu+\xi)B,\\
\frac{dS_m}{dt} &= \xi B - \mu S_m - \lambda_2 S_m,\\
\frac{dE_m}{dt} &= \lambda_2 S_m - (\mu+\gamma_2)E_m,\\
\frac{dI_m}{dt} &= \gamma_2 E_m - (\mu+\kappa)I_m,\\
\frac{dR_m}{dt} &= \kappa I_m - \mu R_m.
\end{cases} \tag{11}$$

The disease-free equilibrium point (DFE) denoted by $E_0 = (C^0, G^0, V_1^0, S_f^0, V_2^0, E_f^0, I_f^0, R_f^0, Cc^0, B^0, S_m^0, E_m^0, I_m^0, R_m^0)$ of the model system (11) is $E_0 = \Gamma_1 \bigoplus \Gamma_2 \bigoplus \Gamma_3$, where $\Gamma_1 = (C^0, G^0, V_1^0, S_f^0)$, $\Gamma_2 = (V_2^0, E_f^0, I_f^0, R_f^0, Cc^0)$ and $\Gamma_3 = (B^0, S_m^0, E_m^0, I_m^0, R_m^0)$ such that

$$\Gamma_1 = \left[\frac{\pi}{\mu+\xi}, \frac{\pi\xi}{(\mu+\xi)\mu}, \frac{\pi\xi\sigma_1}{(\mu+\xi)^2\mu}, \frac{\pi\xi(1-\sigma_1)}{(\mu+\sigma_2)(\mu+\xi)\mu}\right]$$

$$\Gamma_2 = \left[\frac{\pi\xi}{\mu^2(\mu+\xi)}\left(\frac{(1-\sigma_1)\sigma_2}{\mu(\mu+\sigma_2)} + \frac{\sigma_1\xi}{\mu+\xi}\right), 0, 0, 0, 0\right]$$

$$\Gamma_3 = \left[\frac{\pi\xi}{(\mu+\xi)^2}, \frac{\pi\xi^2}{\mu(\mu+\xi)^2}, 0, 0, 0\right]$$

As detailed earlier in Section 5.2, the matrices F and V, representing new infections and the remaining population dynamics, are obtained from the partial derivatives evaluated at the system's equilibrium. Their explicit forms are:

$$F = \left[\frac{\partial \mathcal{F}_i}{\partial(E_f, I_f, Cc, E_m, I_m)}\right]_{E_0} \text{ and } \quad V = \left[\frac{\partial \mathcal{V}_i}{\partial(E_f, I_f, Cc, E_m, I_m)}\right]_{E_0} \tag{12}$$

The model system (11), the infectious compartments are

$$\begin{cases}
\frac{dE_f}{dt} &= (1-\sigma_2)\lambda_1 S_f - (\mu+\gamma_1)E_f,\\
\frac{dI_f}{dt} &= \gamma_1 E_f - (\mu+\rho)I_f,\\
\frac{dCc}{dt} &= (1-\rho)I_f - (\mu+\nu)Cc,\\
\frac{dE_m}{dt} &= \lambda_2 S_m - (\mu+\gamma_2)E_m,\\
\frac{dI_m}{dt} &= \gamma_2 E_m - (\mu+\kappa)I_m.
\end{cases} \tag{13}$$

Now, from Eq. (13) $\mathcal{F}_i$ and $\mathcal{V}_i$ are as follows

$$\mathcal{F}_i = \begin{pmatrix}\mathcal{F}_1\\ \mathcal{F}_2\\ \mathcal{F}_3\\ \mathcal{F}_4\\ \mathcal{F}_5\end{pmatrix} = \begin{pmatrix}(1-\sigma_2)\lambda_1 S_f\\ 0\\ 0\\ \lambda_2 S_m\\ 0\end{pmatrix} = \begin{pmatrix}(1-\sigma_2)\beta_m\frac{I_m}{N_m}S_f\\ 0\\ 0\\ \beta_f\frac{I_f}{N_f}S_m\\ 0\end{pmatrix}, \tag{14}$$

and

$$\mathcal{V}_i = \begin{pmatrix}\mathcal{V}_1\\ \mathcal{V}_2\\ \mathcal{V}_3\\ \mathcal{V}_4\\ \mathcal{V}_5\end{pmatrix} = \begin{pmatrix}(\mu+\gamma_1)E_f\\ (\mu+\rho)I_f - \gamma_1 E_f\\ (\mu+\nu)Cc - (1-\rho)I_f\\ (\mu+\gamma_2)E_m\\ (\mu+\kappa)I_m - \gamma_2 E_m\end{pmatrix}. \tag{15}$$

Evaluating the partial derivatives of $\mathcal{F}_i$ and $\mathcal{V}_i$ at the DFE with respect to E_f, I_f, C_c, E_m, and I_m yields the matrices F and V, which are then used to construct $M_0 = F\lambda^{-1} - V$, given explicitly as follows:

$$M_0 = \begin{pmatrix}
-(\mu+\gamma_1) & 0 & 0 & 0 & \frac{(1-\sigma_2)\beta_m S_f^0}{N_m^0}\lambda^{-1}\\
\gamma_1 & -(\mu+\rho) & 0 & 0 & 0\\
0 & (1-\rho) & -(\mu+\nu) & 0 & 0\\
0 & 0 & \frac{\beta_f S_m^0}{N_f^0}\lambda^{-1} & -(\mu+\gamma_2) & 0\\
0 & 0 & 0 & \gamma_2 & -(\mu+\kappa)
\end{pmatrix}. \tag{16}$$

The matrix $F\lambda^{-1} - V$ encodes the interactions between the different compartments of the system, with each entry representing the rate of transition from one state to another. Its corresponding digraph provides a visual representation of these interactions, where nodes denote the compartments and directed edges indicate the transition rates as defined by the matrix elements.

5.3.1. Matrix and digraph relationship

In the digraph shown in Fig. 14, the nodes E_f, I_f, Cc, E_m, and I_m represent distinct epidemiological states. Directed edges between nodes correspond to non-zero off-diagonal elements of the matrix, capturing the transition rates between compartments, such as γ_1 or $\frac{(1-\sigma_2)\beta_m S_f^0}{N_m^0}\lambda^{-1}$. Self-loops on nodes represent diagonal elements, for example $-(\mu+\gamma_1)$, reflecting exit rates from each compartment due to recovery or natural mortality. Thus, the digraph provides a clear visual mapping of both intra-compartmental exits and inter-compartmental flows within the system.

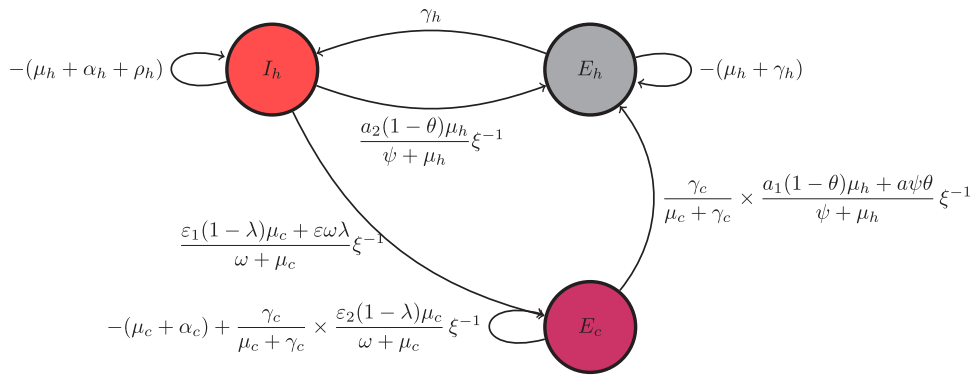


Fig. 10. Digraph representation of $\mathcal{K}\xi^{-1}-\mathcal{B}$ without node I_c.

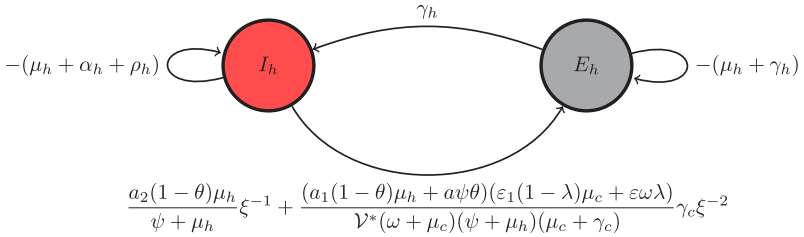


Fig. 11. Digraph representation of $\mathcal{K}\xi^{-1}-\mathcal{B}$ without node E_c and I_c.

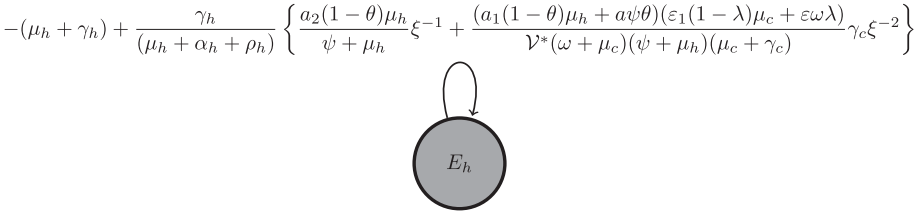


Fig. 12. Digraph representation of $\mathcal{K}\xi^{-1}-\mathcal{B}$ without node I_h, E_c and I_c.

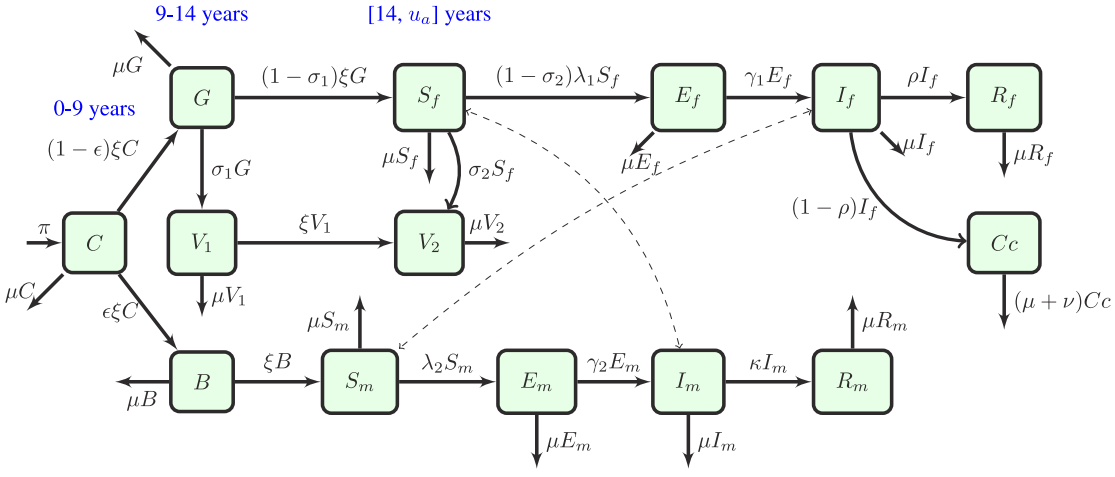


Fig. 13. The schematic flowchart of HPV Vaccination model.

5.3.2. *Digraph reduction and Gaussian elimination*

As shown in Theorem 1, digraph reduction mirrors Gaussian elimination on the matrix $F\lambda^{-1}-V$, removing intermediate nodes by adjusting edge weights until a trivial node, with self-loop -1, is obtained. Following Sections 3 and 4, trivial nodes are created by normalizing their diagonal entries equivalent to row scaling and arcs passing through them are replaced by direct connections between remaining nodes, preserving the determinant. For example, the node I_m has a self-loop $-(\mu+\kappa)<0$, normalized to -1 by dividing incoming arcs

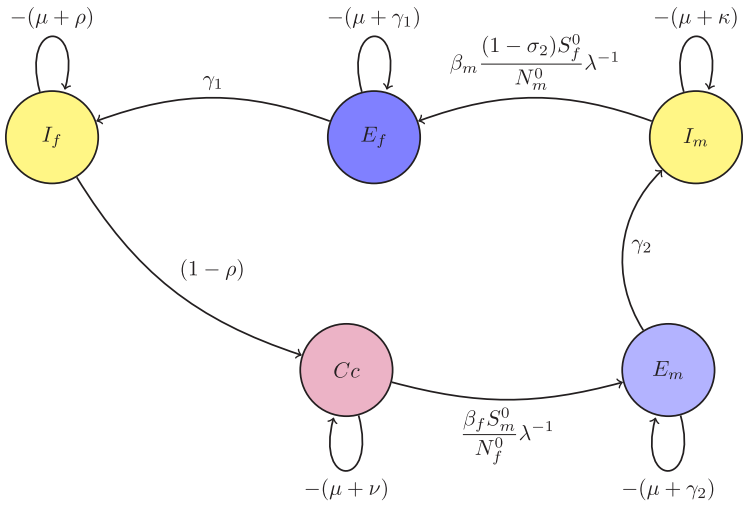


Fig. 14. Digraph representation of $F\lambda^{-1}$ - V.

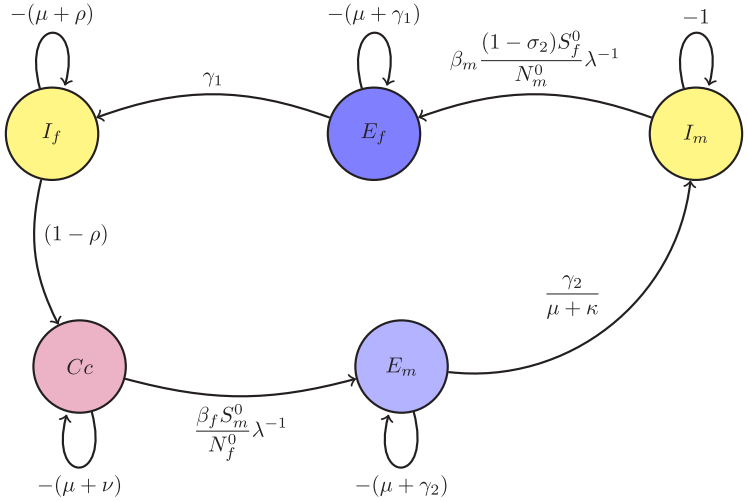


Fig. 15. Creating a trivial node at node I_m.

by $\mu+\kappa$, scaling the corresponding row in $\det(F\lambda^{-1}-V)$. Paths like $E_m \to I_m$ are replaced by direct arcs weighted $\frac{\gamma_2}{\mu+\kappa}$, while nodes without connections can be omitted. Fig. 15 illustrates this process (see Figs. 16, 18 and 19).

The associated matrix is presented

$$M_1 = \begin{pmatrix} -(\mu+\gamma_1) & 0 & 0 & \frac{(1-\sigma_2)\gamma_2\beta_m S_f^0}{N_m^0(\mu+\kappa)}\lambda^{-1} \\ \gamma_1 & -(\mu+\rho) & 0 & 0 \\ 0 & (1-\rho) & -(\mu+\nu) & 0 \\ 0 & 0 & \frac{\beta_f S_m^0}{N_f^0}\lambda^{-1} & -(\mu+\gamma_2) \end{pmatrix}. \quad (17)$$

For the digraph node E_m, its self-loop $-(\mu+\gamma_2)<0$ is normalized to -1 by dividing all incoming arcs by $\mu+\gamma_2$, creating a trivial node. On paths such as $Cc \to E_m \to E_f$, the arc from Cc to E_f is replaced by a direct connection weighted by the product of the original arcs, $\frac{\beta_f S_m^0}{N_f^0(\mu+\gamma_2)}\lambda^{-1}$, as illustrated in Fig. 17.

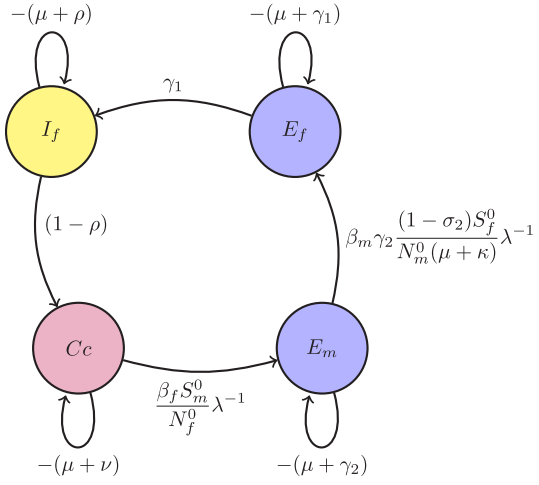


Fig. 16. Digraph representation without node I_m.

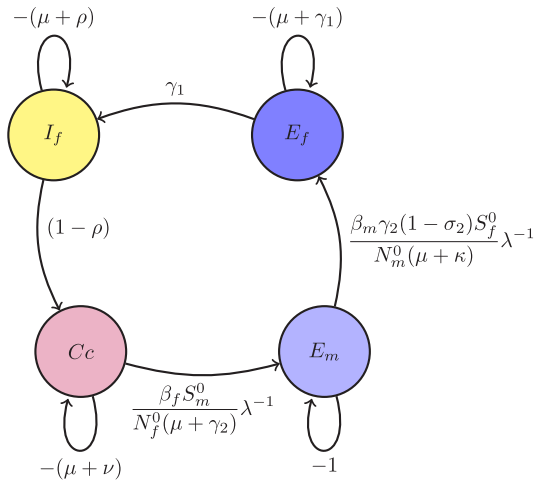


Fig. 17. Creating a trivial node at node E_m.

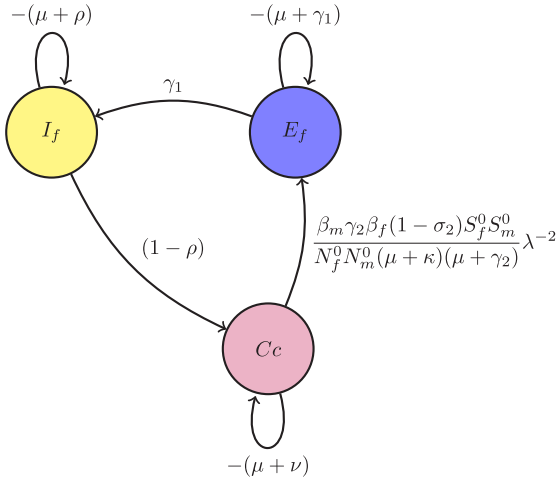


Fig. 18. Digraph Representation without node E_m.

The associated matrix of reduction on digraph without node E_m

$$M_2 = \begin{pmatrix} -(\mu+\gamma_1) & 0 & \dfrac{\beta_f S_m^0(1-\sigma_2)\beta_m S_f^0}{N_f^0N_m^0(\mu+\kappa)(\mu+\gamma_2)}\lambda^{-2} \\ \gamma_1 & -(\mu+\rho) & 0 \\ 0 & (1-\rho) & -(\mu+\nu) \end{pmatrix}. \tag{18}$$

The associated matrix of reduction on digraph without node Cc

$$M_3 = \begin{pmatrix} -(\mu+\gamma_1) & \dfrac{\beta_f S_m^0(1-\sigma_2)(1-\rho)\beta_m S_f^0}{N_f^0N_m^0(\mu+\kappa)(\mu+\gamma_2)(\mu+\nu)}\lambda^{-2} \\ \gamma_1 & -(\mu+\rho) \end{pmatrix}. \tag{19}$$

As in Theorem 1, by setting the self-loop at node E_f to zero, one obtains a polynomial equation in λ^{-2}. Substituting the equilibrium values N_f^0, N_m^0, S_f^0, and S_m^0 and simplifying, the reciprocal of the smallest positive root provides the basic reproduction number $\mathcal{R}_0$ (see Figs. 20–22).

$$\begin{aligned} \lambda^2 &= \frac{(1-\rho)\beta_m\gamma_2\gamma_1\beta_f(1-\sigma_2)S_f^0S_m^0}{(\mu+\kappa)(\mu+\gamma_2)(\mu+\nu)(\mu+\rho)(\mu+\gamma_1)N_f^0N_m^0} \\ \implies \mathcal{R}_0 &= \sqrt{\frac{(1-\rho)\pi\beta_f\beta_m\gamma_2\gamma_1(1-\sigma_2)\mu^2}{(\mu+\kappa)(\mu+\gamma_2)(\mu+\nu)(\mu+\rho)(\mu+\gamma_1)(\mu^2+1)}} \end{aligned} \tag{20}$$

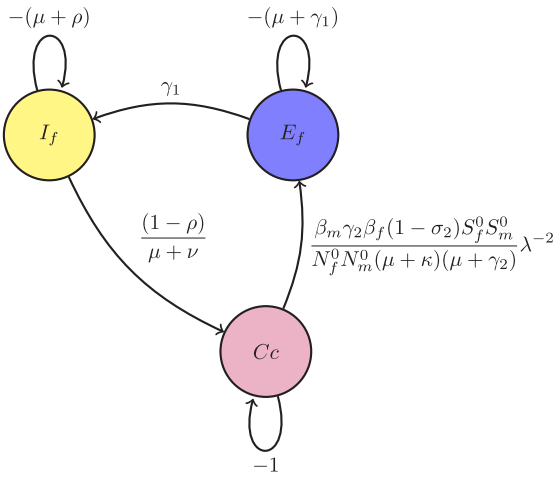


Fig. 19. Creating a trivial node at node Cc.

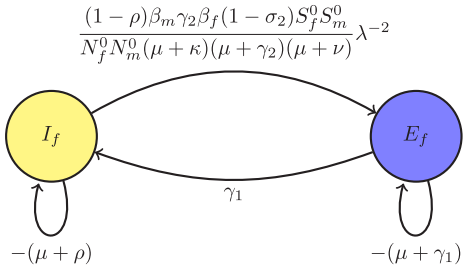


Fig. 20. Digraph Representation without node Cc.

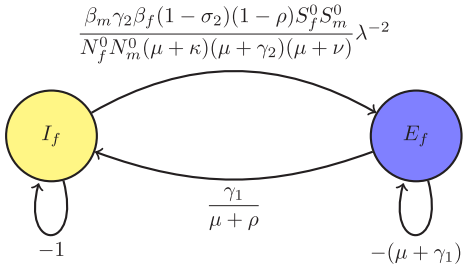


Fig. 21. Creating a trivial node at node I_f.

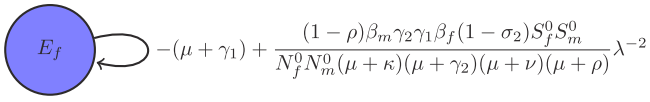


Fig. 22. Digraph representation without node I_f.

5.4. Fractional-order model of HPV spread

Building on the foundational work of Oswald et al. (2025b), this study introduces a Caputo fractional-order HPV transmission model that incorporates memory effects to better capture the persistence of infection. The total population $N(t) = N_f(t) + N_m(t)$ is structured into six female compartments, susceptible S_f, vaccinated V, exposed E_f, infectious I_f, cervical cancer C, and recovered R_f and four male compartments, susceptible S_m, exposed E_m, infectious I_m, and recovered R_m. Recruitment occurs at rates $(1-\phi)\pi_f^\alpha$ into S_f, $\phi\pi_f^\alpha$ into V, and π_m^α into S_m. Transmission arises through effective sexual contact, with forces of infection $\lambda_m^\alpha = \frac{\zeta_{m\to f}^\alpha(\psi_m^\alpha E_m + I_m)}{N_m}$ for females and $\lambda_f^\alpha =$

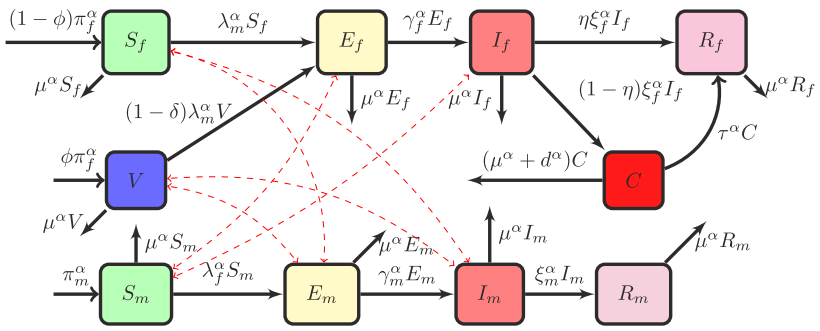


Fig. 23. Transition diagram for the HPV vaccination and treatment.

$\frac{\zeta^\alpha_{f\to m}(\psi^\alpha_f E_f + I_f)}{N_f}$ for males. Susceptible females and males become exposed at rates $\lambda_m^\alpha S_f$ and $\lambda_f^\alpha S_m$, while vaccinated females may be infected at reduced-risk rate $(1-\delta)\lambda_m^\alpha V$ with $0 < \delta < 1$. Exposed individuals progress to infectious classes at rates γ_f^α and γ_m^α. Infected females recover at rate $\eta\xi_f^\alpha$ or progress to cervical cancer at rate $(1-\eta)\xi_f^\alpha$, with cancer cases recovering at rate τ^α or dying at rate $\mu^\alpha + d^\alpha$. Infected males recover at rate ξ_m^α, joining the recovered class R_m (see Figs. 23 and 24).

$$\begin{cases} {}^C_{t_0}D_t^\alpha S_f(t) &= (1-\phi)\pi_f^\alpha - (\mu^\alpha + \lambda_m^\alpha)S_f(t), \\ {}^C_{t_0}D_t^\alpha V(t) &= \phi\pi_f^\alpha - \mu^\alpha V(t) - (1-\delta)\lambda_m^\alpha V(t), \\ {}^C_{t_0}D_t^\alpha E_f(t) &= \lambda_m^\alpha S_f(t) + (1-\delta)\lambda_m^\alpha V(t) - (\mu^\alpha + \gamma_f^\alpha)E_f(t), \\ {}^C_{t_0}D_t^\alpha I_f(t) &= \gamma_f^\alpha E_f(t) - (\mu^\alpha + \xi_f^\alpha)I_f(t), \\ {}^C_{t_0}D_t^\alpha R_f(t) &= \eta\xi_f^\alpha I_f(t) + \tau^\alpha C(t) - \mu^\alpha R_f(t), \\ {}^C_{t_0}D_t^\alpha C(t) &= (1-\eta)\xi_f^\alpha I_f(t) - (\mu^\alpha + d^\alpha + \tau^\alpha)C(t), \\ {}^C_{t_0}D_t^\alpha S_m(t) &= \pi_m^\alpha - (\mu^\alpha + \lambda_f^\alpha)S_m(t), \\ {}^C_{t_0}D_t^\alpha E_m(t) &= \lambda_f^\alpha S_m(t) - (\mu^\alpha + \gamma_m^\alpha)E_m(t), \\ {}^C_{t_0}D_t^\alpha I_m(t) &= \gamma_m^\alpha E_m(t) - (\mu^\alpha + \xi_m^\alpha)I_m(t), \\ {}^C_{t_0}D_t^\alpha R_m(t) &= \xi_m^\alpha I_m(t) - \mu^\alpha R_m(t), \end{cases} \tag{21}$$

In the papillomavirus-free context, the community is free from HPV infection, with the infected class taking a value of zero. Thus, the equilibrium point is given by:

$$PFE = \left(\frac{(1-\phi)\pi_f^\alpha}{\mu^\alpha}, \frac{\phi\pi_f^\alpha}{\mu^\alpha}, 0, 0, 0, 0, \frac{\pi_m^\alpha}{\mu^\alpha}, 0, 0, 0\right)$$

The $\mathbb{GT}$ approach separates new infection terms from other population dynamics. Here, F is an $n \times n$ matrix and V a non-singular $m \times m$ matrix, both derived from partial derivatives at equilibrium.

$$\begin{cases} {}^C_{t_0}D^\alpha E_f(t) &= \lambda_m^\alpha S_f(t) + (1-\delta)\lambda_m^\alpha V(t) - (\mu^\alpha + \gamma_f^\alpha)E_f(t), \\ {}^C_{t_0}D_t^\alpha I_f(t) &= \gamma_f^\alpha E_f(t) - (\mu^\alpha + \xi_f^\alpha)I_f(t), \\ {}^C_{t_0}D_t^\alpha C(t) &= (1-\eta)\xi_f^\alpha I_f(t) - (\mu^\alpha + d^\alpha + \tau^\alpha)C(t), \\ {}^C_{t_0}D_t^\alpha E_m(t) &= \lambda_m^\alpha S_m(t) - (\mu^\alpha + \gamma_m^\alpha)E_m(t), \\ {}^C_{t_0}D_t^\alpha I_m(t) &= \gamma_m^\alpha E_m(t) - (\mu^\alpha + \xi_m^\alpha)I_m(t). \end{cases} \tag{22}$$

Now, from Eq. (22) $\mathcal{F}_i$ and $\mathcal{V}_i$ are as follows

$$\mathcal{F}_i = \begin{pmatrix} \mathcal{F}_1 \\ \mathcal{F}_2 \\ \mathcal{F}_3 \\ \mathcal{F}_4 \\ \mathcal{F}_5 \end{pmatrix} = \begin{pmatrix} \lambda_m^\alpha(S_f(t) + (1-\delta)V(t)) \\ 0 \\ 0 \\ \lambda_f^\alpha S_m(t) \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} \zeta^\alpha_{m\to f}\dfrac{(\psi_m E_m(t) + I_m(t))}{N_m(t)}(S_f + (1-\delta)V(t)) \\ 0 \\ 0 \\ \zeta^\alpha_{f\to m}\dfrac{(\psi_f E_f(t) + I_f(t))}{N_f(t)}S_m(t) \\ 0 \end{pmatrix}, \tag{23}$$

and

$$\mathcal{V}_i = \begin{pmatrix} \mathcal{V}_1 \\ \mathcal{V}_2 \\ \mathcal{V}_3 \\ \mathcal{V}_4 \\ \mathcal{V}_5 \end{pmatrix} = \begin{pmatrix} (\mu^\alpha + \gamma_f^\alpha)E_f(t) \\ (\mu^\alpha + \xi_f^\alpha)I_f(t) - \gamma_f^\alpha E_f(t) \\ (\mu^\alpha + d^\alpha + \tau^\alpha)C(t) - (1-\eta)\xi_f^\alpha I_f(t) \\ (\mu^\alpha + \gamma_m^\alpha)E_m(t) \\ (\mu^\alpha + \xi_m^\alpha)I_m(t) - \gamma_m^\alpha E_m(t) \end{pmatrix}. \tag{24}$$

Now, evaluating the fractional partial derivative of $\mathcal{F}_i$ and $\mathcal{V}_i$ at PFE with respect to E_f, I_f, C, E_m and I_m gives F and V, that is then evaluated to obtain the matrix $Q = F\lambda^{-1} - V$ as follows:

$$Q = \begin{pmatrix} -(\mu^\alpha + \gamma_f^\alpha) & 0 & 0 & 0 & \frac{\zeta^\alpha_{m\to f}(2-\phi-\delta)\pi_f^\alpha}{\pi_m^\alpha}\lambda^{-1} \\ \gamma_f^\alpha & -(\mu^\alpha + \xi_f^\alpha) & 0 & 0 & 0 \\ 0 & (1-\eta)\xi_f^\alpha & -(\mu^\alpha + d^\alpha + \tau^\alpha) & 0 & 0 \\ \frac{\zeta^\alpha_{f\to m}\pi_m^\alpha}{\pi_f^\alpha}\lambda^{-1} & \frac{\zeta^\alpha_{f\to m}\pi_m^\alpha}{\pi_f^\alpha}\lambda^{-1} & \frac{\zeta^\alpha_{f\to m}\pi_m^\alpha}{\pi_f^\alpha}\lambda^{-1} & -(\mu^\alpha + \gamma_m^\alpha) & 0 \\ 0 & 0 & 0 & \gamma_m^\alpha & -(\mu^\alpha + \xi_m^\alpha) \end{pmatrix}. \tag{25}$$

5.5. Illustration of the matrix and digraph relationships in the model

The matrix Q encodes the transition rates between epidemiological compartments, with each entry representing movement between states such as exposed, infected, and recovered. Its digraph representation depicts these interactions: nodes correspond to compartments (E_f, I_f, C, E_m, I_m), and directed edges represent transition rates, e.g., γ_f^α or $\frac{\zeta^\alpha_{m\to f}(2-\phi-\delta)\pi_f^\alpha}{\pi_m^\alpha}$. Self-loops denote internal transitions like recovery or natural death, while off-diagonal entries capture flows between compartments, as shown

5.5.1. Digraph reduction and Gaussian elimination

Digraph reduction parallels Gaussian elimination on $F\lambda^{-1} - V$, simplifying the system by removing intermediate nodes, similar to eliminating off-diagonal elements. A node becomes trivial when its self-loop is normalized to -1, reflecting constant rates such as natural mortality. The process follows two steps: (1) normalize incoming edges to create a trivial node, and (2) replace paths through it with direct connections, preserving the determinant (Oswald et al., 2025c). For instance, node I_m is reduced by dividing its incoming edges by $(\mu^\alpha + \xi_m^\alpha)$, with connections updated accordingly, as shown in Fig. 25.

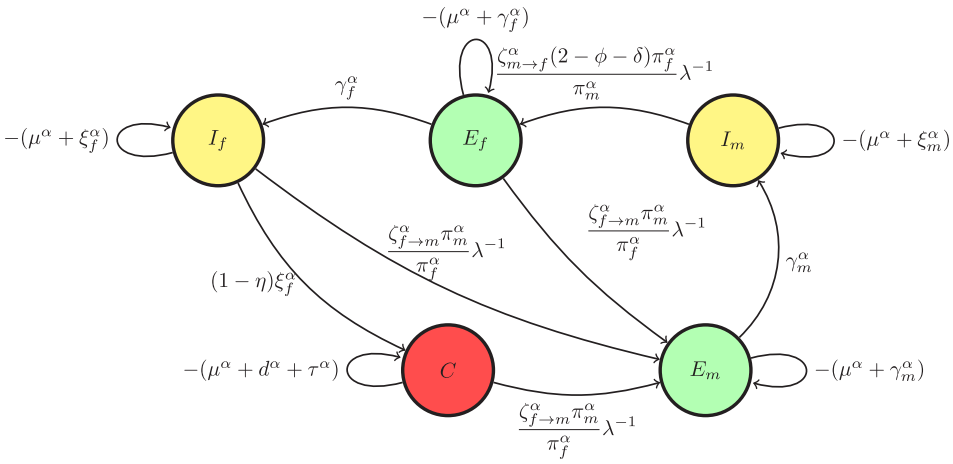


Fig. 24. Digraph representation of $F\lambda^{-1}-V$.

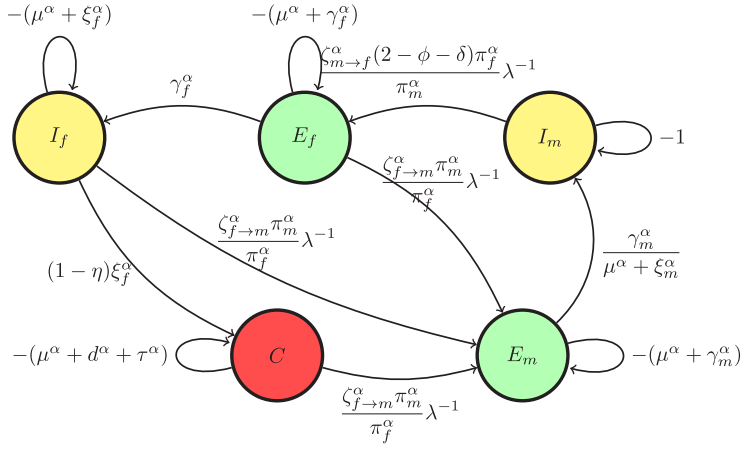


Fig. 25. Creating a trivial node at node I_m.

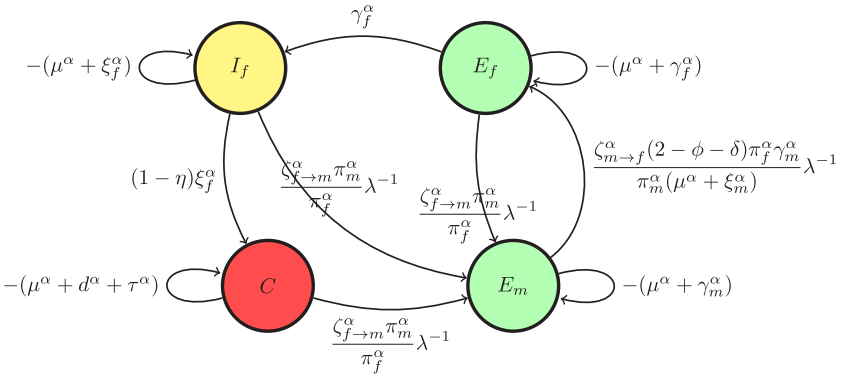


Fig. 26. Digraph representation without node I_m.

The associated matrix is presented

$$M_1 = \begin{pmatrix} -(\mu^\alpha+\gamma_f^\alpha) & 0 & 0 & \frac{\zeta^\alpha_{m\to f}(2-\phi-\delta)\pi_f^\alpha\gamma_m^\alpha}{\pi_m^\alpha(\mu^\alpha+\xi_m^\alpha)}\lambda^{-1} \\ \gamma_f^\alpha & -(\mu^\alpha+\xi_f^\alpha) & 0 & 0 \\ 0 & (1-\eta)\xi_f^\alpha & -(\mu^\alpha+d^\alpha+\tau^\alpha) & 0 \\ \frac{\zeta^\alpha_{f\to m}\pi_m^\alpha}{\pi_f^\alpha}\lambda^{-1} & \frac{\zeta^\alpha_{f\to m}\pi_m^\alpha}{\pi_f^\alpha}\lambda^{-1} & \frac{\zeta^\alpha_{f\to m}\pi_m^\alpha}{\pi_f^\alpha}\lambda^{-1} & -(\mu^\alpha+\gamma_m^\alpha) \end{pmatrix}. \tag{26}$$

Reducing node E_m involves normalizing its self-loop $-(\mu^\alpha+\gamma_m^\alpha)$ by dividing all incoming arcs by $\mu^\alpha+\gamma_m^\alpha$, creating a trivial node with loop weight -1. The resulting reduced matrix is(see Box II).

The Gaussian elimination procedure for all other nodes in Figs. 28–29 is carried out in the same manner as illustrated in Figs. 26 and 27, ultimately leading to the resulting reduced Fig. 29.

Finally, following the procedure outlined in Theorem 1, by setting the weight of the self-loop at node E_f to zero, we obtain the following

$$M_2 = \begin{pmatrix} -(\mu^\alpha + \gamma_f^\alpha) + \dfrac{\zeta_{f\to m}^\alpha \zeta_{m\to f}^\alpha (2-\phi-\delta)\gamma_m^\alpha}{(\mu^\alpha + \xi_m^\alpha)(\mu^\alpha + \gamma_m^\alpha)}\lambda^{-2} & 0 & \dfrac{\zeta_{f\to m}^\alpha \zeta_{m\to f}^\alpha (2-\phi-\delta)\gamma_m^\alpha}{(\mu^\alpha + \xi_m^\alpha)(\mu^\alpha + \gamma_m^\alpha)}\lambda^{-2} \\ \gamma_f^\alpha & -(\mu^\alpha + \xi_f^\alpha) & 0 \\ 0 & (1-\eta)\xi_f^\alpha & -(\mu^\alpha + d^\alpha + \tau^\alpha) \end{pmatrix}.$$

Box II.

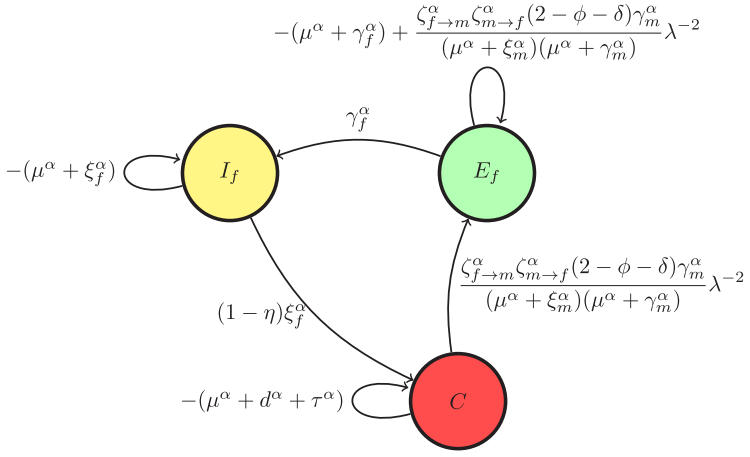


Fig. 27. Digraph representation without node E_m.

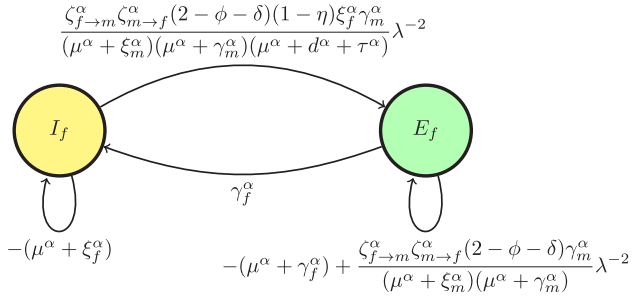


Fig. 28. Digraph representation without node C.

polynomial equation in λ:

$$-(\mu^\alpha + \gamma_f^\alpha) + \frac{\zeta_{f\to m}^\alpha \zeta_{m\to f}^\alpha (2-\phi-\delta)\gamma_m^\alpha}{(\mu^\alpha + \xi_m^\alpha)(\mu^\alpha + \gamma_m^\alpha)}\left(1 + \frac{(1-\eta)\xi_f^\alpha \gamma_f^\alpha}{(\mu^\alpha + \xi_f^\alpha)(\mu^\alpha + d^\alpha + \tau^\alpha)}\right)\lambda^{-2} = 0 \tag{27}$$

The reciprocal of the smallest positive root, say λ of the polynomial Eq. (27), gives the expression for $\mathcal{R}_e$.

$$\begin{aligned} \lambda^2 &= \frac{\zeta_{f\to m}^\alpha \zeta_{m\to f}^\alpha (2-\phi-\delta)\gamma_m^\alpha}{(\mu^\alpha + \gamma_f^\alpha)(\mu^\alpha + \xi_m^\alpha)(\mu^\alpha + \gamma_m^\alpha)}\left(1 + \frac{(1-\eta)\xi_f^\alpha \gamma_f^\alpha}{(\mu^\alpha + \xi_f^\alpha)(\mu^\alpha + d^\alpha + \tau^\alpha)}\right) \\ \implies \mathcal{R}_e &= \sqrt{\frac{\zeta_{f\to m}^\alpha \zeta_{m\to f}^\alpha (2-\phi-\delta)\gamma_m^\alpha}{(\mu^\alpha + \gamma_f^\alpha)(\mu^\alpha + \xi_m^\alpha)(\mu^\alpha + \gamma_m^\alpha)}\left(1 + \frac{(1-\eta)\xi_f^\alpha \gamma_f^\alpha}{(\mu^\alpha + \xi_f^\alpha)(\mu^\alpha + d^\alpha + \tau^\alpha)}\right)} \end{aligned} \tag{28}$$

After understanding the graph-theoretic framework for computing the basic and effective reproduction numbers as presented above, interested readers may consult recent studies that apply similar approaches to fractional-order epidemic models for rabies and norovirus. These works (Ndendya and Msigwa, 2026; Ndendya et al., 2025c,b,a,e) demonstrate how digraph or graph-based methods can be used to compute $\mathcal{R}_0$ and $\mathcal{R}_e$, providing opportunities to verify computations, explore the impact of different fractional derivatives, and gain deeper insights into the dynamics and control of complex epidemics. By studying these examples, readers can appreciate the versatility of graph-theoretic reductions and their potential for both analytical and numerical exploration of high-dimensional or memory-dependent epidemic systems.

6. Conclusion

We introduce a graph-theoretic digraph reduction framework for computing the basic and effective reproduction numbers, $\mathcal{R}_0$ and $\mathcal{R}_e$, in continuous-time and fractional-order epidemic models. Applicable to compartmental systems with a disease-free equilibrium (DFE), the approach linearizes the infected subsystem and partitions the dynamics into infection and transition components, yielding the classical next-generation matrix structure. Rather than directly computing matrix inverses or eigenvalues, the method systematically reduces the associated Coates digraph, a process mathematically equivalent to symbolic Gaussian elimination. This procedure enables efficient computation of $\mathcal{R}_0$ and $\mathcal{R}_e$, provides tight lower bounds, and explicitly identifies dominant transmission pathways and compartmental contributions, thereby enhancing interpretability, numerical stability, and visualization clarity.

Applications to a variety of ordinary differential equation (ODE) and fractional-order epidemic models demonstrate the practical advantages of the framework. The digraph reduction method preserves accuracy while substantially reducing computational complexity compared with traditional next-generation matrix approaches, particularly for high-dimensional or network-structured systems. In fractional-order models, it naturally accommodates memory effects, capturing threshold dynamics without compromising interpretability. Overall, the framework offers a transparent, modular, and scalable tool for both quantitative evaluation of intervention strategies and qualitative understanding of transmission structures.

Despite the demonstrated strengths of the digraph reduction framework, several important directions remain for future research. In particular, further work is needed to extend the approach to settings where

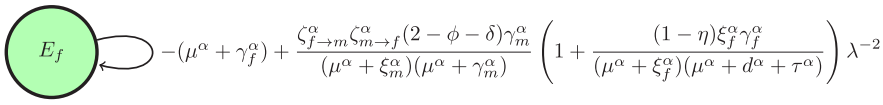


Fig. 29. Digraph representation without node I_f.

a well-defined disease-free equilibrium (DFE) or a clearly linearized infected subsystem is not readily available, as often occurs in highly stochastic, adaptive, or data-driven epidemic models. Expanding the methodology to fully stochastic and time-varying networked systems would significantly broaden its applicability and allow for the analysis of more realistic epidemic scenarios.

CRediT authorship contribution statement

Sylas Oswald: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Eunice Mureithi:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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